

Electrical Enclosure Heat Transfer Analysis

TTM F2B Measuring Machine Capstone Group 1407

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Objective

This analysis is used to determine the internal heat generation (Q_I) and airflow required (CFM) for the electrical enclosure to ensure all internal components remain within their maximum specified operating temperatures.

Givens

The following active components are assumed to operate simultaneously with maximal heat generation, determining the total heat load. The heat dissipation for each component is calculated using maximum power draw or operating efficiency.

Table 1: Component Heat Dissipation (Q_I)

Component	Input/Capacity	Heat Loss Calculation	Heat Dissipation (Q)
PC (Edge Gateway)	60.0 W	Assumed 100 % Loss (Worst-Case)	60.0 W
Ethernet Switch	1.2 W	Assumed 100 % Loss (Worst-Case)	1.2 W
Power Supply (PSU)	180 W (Input)	87 % Efficient ($180 \text{ W} \times 0.13$)	23.4 W
Analog Output Module	6.0 W	Assumed 100 % Loss ($30 \text{ V} \times 0.2 \text{ A}$)	6.0 W
Total Internal Heat Dissipation (Q_I)			90.6 W

- Density of air ($\rho = 1.225 \text{ kg m}^{-3}$)
- Specific heat of air ($c_p = 1.005 \text{ kJ kg}^{-1} \text{ K}^{-1}$).

Note:

- Components specified with 100 % loss (PC, Ethernet Switch, Analog Module) are assumed to convert their entire input power draw directly into heat, representing the worst-case scenario.
- Power Supply heat dissipation is calculated based on its inefficiency ($100 \% - 87 \% = 13 \% \text{ loss}$) applied to the 180 W input capacity, resulting in 23.4 W of heat lost.

Assumptions

The analysis is based on worst-case parameters. These values are used to calculate fan capacity.

- **Maximum Power Assumption:** All active components are assumed to be operating at their maximum current/voltage draw simultaneously.
- **Maximum Internal Temperature:** The air temperature inside the enclosure cannot exceed 40°C (Analog Output Module).

- **Internal Heat Generation:** All heat generators are considered one combined heat source (Q_I).
- **Forced Convection Dominance:** The use of a cooling fan means forced convection is the dominant heat transfer mechanism.
- **Negligible Back Heat Transfer:** Heat transfer through enclosure walls is assumed to be negligible compared to forced convection.
- **Ambient Temperature:** The ambient factory temperature is $\approx 21^\circ\text{C}$ (70°F).
- **Internal Temperature:** A 30°C internal box temperature is considered sufficient for component cooling.

Electrical Box Control Volume

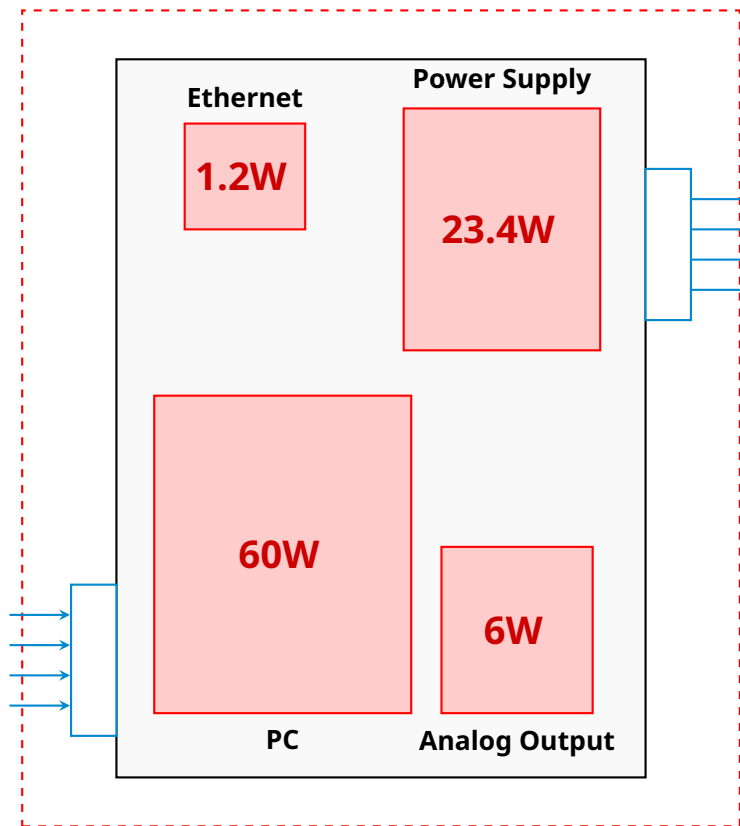


Figure 1: Control Volume Diagram of the Electrical Enclosure.

The control volume defined in Figure 1 encompasses the internal air space of the enclosure, where the boundary accounts for the energy balance between the internal heat generation (Q_I) and the mass flow of air.

Thermal Management Goal

The thermal management goal is to remove an internal heat load of $Q_I = 90.6\text{ W}$, drawing air in at $T_{Ambient} = 21^\circ\text{C}$ and exhausting it at $T_{Exhaust} = 30^\circ\text{C}$ utilizing forced convection.

Calculations

Mass Flow Rate ($\dot{m}$)

$$\text{Formula: } \dot{m} = \frac{Q_I}{c_p \times \Delta T}$$

$$\text{Values: } Q_I = 90.6 \text{ W}$$

$$c_p = 1005 \text{ J kg}^{-1} \text{ K}^{-1}$$

$$\Delta T = 9 \text{ K}$$

$$\text{Calculation: } \dot{m} = \frac{90.6 \text{ W}}{1005 \times 9} \approx 0.0100 \text{ kg/s}$$

Volumetric Flow Rate ($\dot{V}$)

$$\text{Formula: } \dot{V} = \frac{\dot{m}}{\rho}$$

$$\text{Calculation: } \dot{V} = \frac{0.0100 \text{ kg s}^{-1}}{1.225 \text{ kg m}^{-3}} \approx 0.0082 \text{ m}^3/\text{s}$$

Volumetric Flow Rate Conversion (CFM)

$$\text{Airflow (CFM)}_{\text{Required}} = \dot{V} (\text{m}^3/\text{s}) \times 2118.88$$

$$\text{Airflow (CFM)}_{\text{Required}} \approx 17.89 \text{ CFM}$$

Conclusion

The minimum required airflow for the fan to successfully remove 90.6 W of internal heat while maintaining the enclosure air temperature below 30 °C is 17.89 CFM.